List of Symbols

Δ	Variation of a function
$\boldsymbol{\Delta}$	Vectorial variation of a vector
d	Ordinary differentiation of a function
$\boldsymbol{d}$	Vectorial differentiation of a vector
$\vec{\nabla}$ or grad	Gradient operator
$\vec{\nabla}\cdot$ or div	Divergence operator
$\vec{\nabla}\times$ or curl	Curl or rotation operator
∇^2	Laplacian operator
$\Box$	d'Alembertian operator
$\boldsymbol{D}$	Covariant differentiation of a tensor
∇	Covariant differentiation of a tensor component
$\boldsymbol{\nabla}$	Total covariant differentiation of a tensor component
$\cdot$	Inner product
$\wedge$	Outer or exterior or wedge product
i_X or $X\rfloor$	Interior product with respect to the vector X
$\pounds_X$	Lie differentiation with respect to the vector X
$\mathbf{d}$	Exterior differentiation
$\mathbf{D}$	Covariant exterior differentiation
$\vec{e}_i$ or e_i	Basis vector
$\widetilde{e}_i$ or ϑ^i	Basis covector/dual vector or one-form
X	Matrix
$\vec{X}$	Four-dimensional vector or three-dimensional vector in $\mathbb{R}^3$
$\boldsymbol{X}$	Four vector
$\widetilde{X}$	Covector
X	Tensor
$X^{i_1}{}_{j_1 j_2}{}^{i_2\cdots i_r}{}_{j_3\cdots j_s}$	Component of a (r,s)-type tensor X
δ_{ij}	Kronecker delta symbol or Euclidean metric tensor
η_{ij}	Minkowski metric tensor
g_{ij}	Riemann metric tensor
$\delta^{\mu_1\mu_2\cdots\mu_r}_{\nu_1\nu_2\cdots\nu_r}$	Generalized Kronecker delta symbol
ε_{ijk}	Three-dimensional Levi–Civita symbol
$\varepsilon_{ijk\ell}$	Four-dimensional Levi–Civita symbol
$\boldsymbol{\eta}_{ijk}$	Three-dimensional Levi–Civita tensor
$\boldsymbol{\eta}_{ijk\ell}$	Four-dimensional Levi–Civita tensor
$\boldsymbol{\eta}$	Volume form
$\left\{ {i \atop jk} \right\}$	Christoffel symbol

$\Gamma^i{}_{jk}$		Affine connection
$\omega^i{}_{jk}$		Linear connection
$R^i{}_{jk\ell}$		Riemann–Christoffel curvature tensor
$C^i{}_{jk\ell}$		Weyl tensor
$T^i{}_{jk}$		Torsion tensor
$K^i{}_{jk}$		Contortion tensor
Q_{ijk}		Non-metricity tensor
$N^i{}_{jk}$		Distortion tensor
$\boldsymbol{d}\mathbf{p}$	or $\mathbf{dp}$	Infinitesimal displacement vector of a point p
$\Gamma^i{}_j$		Affine connection form
$\omega^i{}_j$		Linear connection form
$\Omega^i{}_j$		Curvature two-form
Θ^i		Torsion two-form